

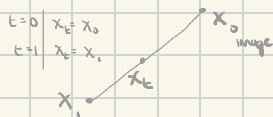
RECTIFIED FLOW

Let $x_0 \in \mathbb{R}^d$ be a flattened image vector st $x_0 \sim \text{Data}$ and $x_1 \in \mathbb{R}^d$, $x_1 \sim N(0, I)$ be the corresponding vector from a base distribution assumed Gaussian

$$x_t = (1-t)x_0 + tx_1, \quad t \in [0, 1]$$

Objective $\rightarrow$ learn a velocity field

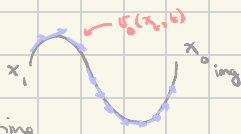
$$\frac{dx}{dt} = v_\theta(x_t, t)$$



$$\mathcal{L}(\theta) = \mathbb{E}_{x_0, x_1, t} [\|v_\theta(x_t, t) - v^*(x_t, t)\|^2]$$

$$\text{where } v^*(x_t, t) = x_1 - x_0$$

$$v^* = \frac{d}{dt} x_t = \frac{d}{dt} [(1-t)x_0 + tx_1] = -x_0 + x_1 = x_1 - x_0$$



$v^* = x_1 - x_0$ is constant wrt time t

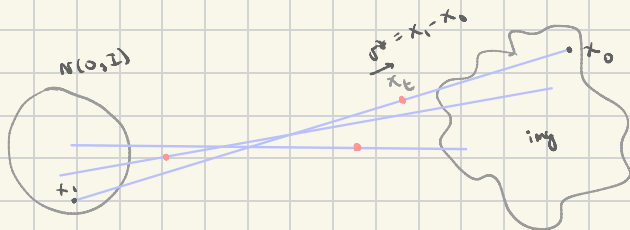
v_θ is a learned vector field across all space-time

- at time t if I see something like this x_t that probably lies on a transport path from a certain kind of noise to a certain kind of image, and here's the direction I should move it in

- It generalises the pattern of motion that connects the entire noise dist to the entire data dist

During prediction, the model doesn't know where it's ultimately going, it only knows how the world tends to move from noise toward structure.

It learned that from seeing many pairs during training



Inference

Solve the ODE

During inference, numerical integration, we multiply that velocity by a tiny step size Δt

$$x_{t+\Delta t} = x_t + v_\theta(x_t, t) \Delta t \quad (A)$$

How to get (A)?

$$\frac{dx}{dt} = v_\theta(x, t)$$

integrate over small interval $[t, t + \Delta t]$

$$x_{t+\Delta t} = x_t + \int_t^{t+\Delta t} v_\theta(x_s, s) ds$$

$$\Delta t = t_{k+1} - t_k$$

Euler approximation (Left Riemann)

$$x_{t+\Delta t} \approx x_t + v_\theta(x_t, t) \Delta t \quad (A)$$